

AIIMS CRE OT Technician

Memory Based Non-Technical PYQ Solution | Part 5

Quantitative Aptitude

Q1. A student's marks increase from 400 to 500. What is the percentage increase?

- A) 20%
- B) 25%
- C) 30%
- D) 50%

Answer: B) 25%

Explanation: Increase in marks = $500 - 400 = 100$.

Percentage Increase = $(\text{Increase} / \text{Original Value}) \times 100 = (100 / 400) \times 100 = 25\%$.

Q2. A shopkeeper buys an item for ₹800 and sells it for ₹920. Find the profit percentage.

- A) 12%
- B) 15%
- C) 18%
- D) 20%

Answer: B) 15%

Explanation: Cost Price (CP) = ₹800. Selling Price (SP) = ₹920.

Profit = $SP - CP = 920 - 800 = ₹120$.

Profit % = $(\text{Profit} / \text{CP}) \times 100 = (120 / 800) \times 100 = 15\%$.

Q3. The ratio of boys to girls in a class is 3:2. If there are 30 boys, how many girls are there?

- A) 15
- B) 18
- C) 20
- D) 25

Answer: C) 20

Explanation: Let the number of girls be x . The ratio is $3/2 = 30/x$.
Solving for x : $3x = 60 \rightarrow x = 20$.

Q4. Find the average of 12, 18, 20, 25, 30.

- A) 20
- B) 20.5
- C) 21
- D) 22

Answer: C) 21

Explanation: Average = (Sum of all values) / (Number of values).
Sum = $12 + 18 + 20 + 25 + 30 = 105$.
Average = $105 / 5 = 21$.

Q5. 'A' can complete a work in 10 days. How many days will it take to complete 50% of the work?

- A) 2 Days
- B) 4 Days
- C) 5 Days
- D) 6 Days

Answer: C) 5 Days

Explanation: If 100% of the work is completed in 10 days, 50% (which is half the work) will take half the time. $10 / 2 = 5$ days.

Q6. A car travels 120 km in 2 hours. What is the speed?

- A) 50 km/h
- B) 60 km/h
- C) 70 km/h
- D) 80 km/h

Answer: B) 60 km/h

Explanation: Speed = Distance / Time.
Speed = $120 \text{ km} / 2 \text{ hours} = 60 \text{ km/h}$.

Q7. Find the simple interest on ₹5000 at 10% per annum for 2 years.

- A) ₹500
- B) ₹800
- C) ₹1000
- D) ₹1200

Answer: C) ₹1000

Explanation: Simple Interest (SI) = $(\text{Principal} \times \text{Rate} \times \text{Time}) / 100$.
 $SI = (5000 \times 10 \times 2) / 100 = 1000$.

Reasoning

Q8. Analogy: Doctor : Hospital :: Teacher : ?

- A) Office
- B) School
- C) Library
- D) Market

Answer: B) School

Explanation: A doctor's primary workplace is a hospital. Similarly, a teacher's primary workplace is a school.

Q9. Choose the odd one out:

- A) Crow
- B) Sparrow
- C) Eagle
- D) Shark

Answer: D) Shark

Explanation: Crow, Sparrow, and Eagle are all birds, whereas a Shark is a fish.

Q10. Complete the number series: 2, 6, 12, 20, ?

- A) 24
- B) 30
- C) 32
- D) 36

Answer: B) 30

Explanation: The difference between consecutive numbers increases by 2 each time:

$$6 - 2 = 4$$

$$12 - 6 = 6$$

$$20 - 12 = 8$$

The next difference will be 10. Therefore, $20 + 10 = 30$.

Q11. If CAT is coded as 3120, then DOG will be coded as?

- A) 4157
- B) 4147
- C) 3157
- D) 4156

Answer: A) 4157

Explanation: Letters are coded according to their alphabetical position (A=1, B=2 ... Z=26).

C=3, A=1, T=20 (3120).

D=4, O=15, G=7 (4157).

Q12. Father's brother is your?

- A) Maternal Uncle
- B) Paternal Uncle
- C) Grandfather
- D) Cousin

Answer: B) Paternal Uncle

Explanation: The brother of one's father is known as a paternal uncle.

Q13. If you face North and turn 90° clockwise, which direction will you face?

- A) South
- B) West
- C) East
- D) South-East

Answer: C) East

Explanation: Rotating 90 degrees clockwise from the North direction lands you pointing towards the East.

Q14. Ravi is 3rd from the left and 2nd from the right in a row. How many students are there in the row?

- A) 4
- B) 5
- C) 6
- D) 7

Answer: A) 4

Explanation: Total students = (Position from Left) + (Position from Right) - 1.
Total = $3 + 2 - 1 = 4$.

General Knowledge

Q15. Which is the world's largest country by electricity production?

- A) United States
- B) India
- C) Russia
- D) China

Answer: D) China

Explanation: China is the largest producer of electricity globally, followed by the United States and India.

Q16. How many floors are there in Burj Khalifa?

- A) 154
- B) 160
- C) 163
- D) 165

Answer: C) 163

Explanation: The Burj Khalifa, located in Dubai, has 163 floors above ground.

Q17. ISRO is the space agency of which country?

- A) Russia
- B) Japan
- C) India
- D) Israel

Answer: C) India

Explanation: ISRO (Indian Space Research Organisation) is the national space agency of India.

English Language

Q18. Choose the correct sentence:

- A) He go the school daily
- B) He goes to school daily
- C) He going to school daily
- D) He go to school daily

Answer: B) He goes to school daily

Explanation: When using the third-person singular pronoun "He" in the simple present tense, the verb must be in its singular form ("goes"). Also, the preposition "to" indicates direction.

Q19. Choose the synonym of 'Happy'.

- A) Sad
- B) Angry
- C) Joyful
- D) Tired

Answer: C) Joyful

Explanation: 'Joyful' has a similar meaning to 'Happy'.

Q20. Identify the correct sentence:

- A) She don't like tea
- B) She doesn't likes tea
- C) She doesn't like tea
- D) She don't likes tea

Answer: C) She doesn't like tea

Explanation: With the third-person singular subject "She", we use the auxiliary verb "does not" (doesn't) combined with the base form of the main verb ("like").